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T11

EXPERIMENT-2

AIM

To understand Version Control System / Source Code Management, install Git and create a GitHub account.

THEORY

What Is Version Control?

Version control, sometimes referred to as source control or revision control, is a system that keeps track of changes made to a file or group of files over time so that you may retrieve particular versions later. Although it can be applied to any circumstance where several versions of something are made and may need to be monitored and recalled, it is most frequently employed in software development.

What is Git?

- Git is a free and open-source distributed version control system designed to handle everything from small to very large projects with speed and efficiency.
- Git relies on distributed development, where more than one developer may have access to the source code of an application and can modify changes that may be seen by other developers.
- Initially designed and developed by Linus Torvalds for Linux kernel development in 2005.
- Every Git working directory is a full-fledged repository with complete history and full version tracking capabilities, independent of network access or a central server.
- Git allows a team of people to work together using the same files, helping them cope with confusion when multiple people edit the same files.

What Is GitHub?

GitHub is a web-based platform that hosts software development projects and uses Git for version management. It helps developers work together on the same projects and keep track of changes made to their code.

GitHub offers:

- A user-friendly interface
- Collaborative tools
- Project management features

Developers can create and manage code in repositories stored remotely, making it accessible to others. GitHub is essentially a collection of repositories containing project files.

Use of Version Control Software

- Allows the user to have “versions” of a project, showing changes made over time and enabling backtracking if necessary.
- Makes it possible to compare two versions or reverse changes, which is invaluable for larger projects.
- Changes are saved as patch files that can be applied to one version to make it the same as the next.
- All versions are stored on a central server, and developers check out and upload changes back to this server.

Use Cases of GitHub

- Version Control: Developers can revert to previous versions if changes affect the entire code.
- Collaboration and Code Review: Teams can review each other’s code and work together efficiently.
- Issue Tracking: Issues can be assigned to specific developers for resolution.
- Open Source Development: GitHub is the most widely used platform for open-source projects.

Characteristics of Git

A. Strong support for non-linear development

- Supports rapid branching and merging.
- Includes tools for visualizing and navigating non-linear history.
- Branches are lightweight.

B. Distributed development

- Each developer has a local copy of the entire history.
- Changes can be merged efficiently.

C. Compatibility with existing systems/protocols

- Git has CVS server emulation, enabling use of existing CVS clients and IDE plugins.

D. Efficient handling of large projects

- Git is fast and scalable compared to other systems.
- Fetching from a local repository is faster than from a remote server.

E. Data Assurance

- Git history is stored so that each version ID depends on the complete development history.
- Old versions cannot be changed without detection.

F. Automatic Garbage Collection

- Git performs garbage collection when enough loose objects exist.
- Garbage collection can be called explicitly using .

G. Periodic explicit object packing

- Git stores each object as a separate file.
- Uses packfiles to store many objects together, delta-compressed.
- Index files specify offsets of objects in packfiles.
- Packing is computationally expensive, so Git defers it when possible.
- Git does periodic repacking automatically, but manual repacking can be done with .

How Does Git Work?

- A Git repository is a key-value object store indexed by SHA-1 hash values.
- Commits, files, tags, and filesystem tree nodes are stored as objects.

- The repository functions like a large hash table with no hash collisions.
- Git works by taking snapshots of files

Installation of GIT